

WASP-135b

R-0136

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-135_241002 · created 2026-07-16T06:49:21+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2460585.6592 +/- 0.0095 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.158 +/- 0.047
Transit depth (Rp/Rs)^2	2.49 +/- 1.48 [%]
Orbital Inclination (inc)	84.25 +/- 2.45
Airmass coefficient 1 (a1)	0.995 +/- 0.031
Airmass coefficient 2 (a2)	0.002 +/- 0.017
Scatter in the residuals of the lightcurve fit is	3.13 %
Best Comparison Star	#2 - [162, 256]
Optimal Aperture	2.82
Optimal Annulus	14.17
Transit Duration (day)	0.08 +/- 0.014

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.158 ± 0.047 vs 0.139 (deviation 0.38σ)
Duration fit vs published	0.08 d vs 0.0687 d (deviation 0.76σ)
Mid-time O–C	0.9 min
Depth SNR	3.2
Residual scatter	3.13 %

Light curve

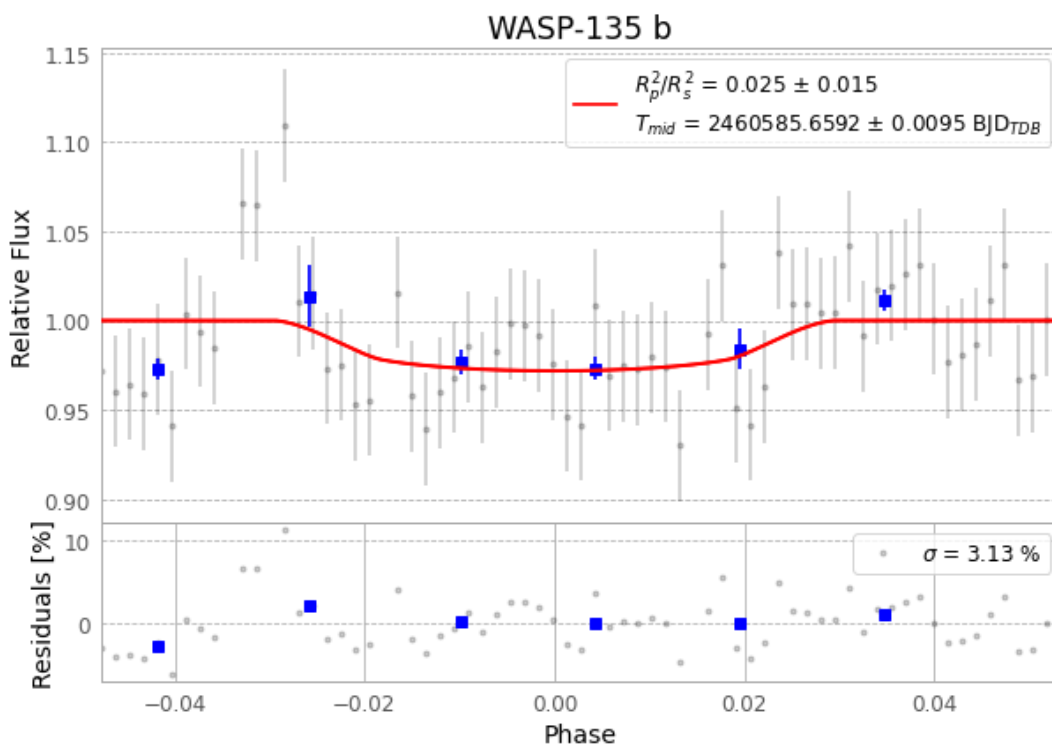


Figure 1 — FinalLightCurve_WASP-135 b_2024-10-01.png (SHA-256 b06f1b3733802ddc...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [318, 255], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 924cbe3e24c779a0...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 3ebc8ba

Data provenance

69 FITS frames (45 MB), WASP-135241002021215.FITS → WASP-135241002053615.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-135-241002.log (cfeb183deeee...), FinalLightCurve_WASP-135 b_2024-10-01.png (b06f1b373380...), FinalLightCurve_WASP-135 b_2024-10-01.csv (3c1fd07abbbb...)

Compute resources

Wall time 305.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-16 06:49Z.